MIND MATRIX

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SESSION: 07

REPORT ON: TASK REPORT

TASK NO: 03

TITLE:

COMPREHENSIVE FEATURE DESIGN & IMPLEMENTATION STRATEGY

Feature Description

Feature: Bill Split & Tip Tracker

This feature allows users to enter the bill amount, number of people, and tip percentage. Users can choose to round up the tip and save previous calculations in a history list.

State Model

State Variable	Type	Initial Value	Description
billAmount	String	""	Stores bill input
numPeople	Int	1	Number of people
tipPercent	Int	15	Tip percentage
roundUp	Boolean	false	Round tip
result	Double	Derived	Calculated tip
historyList	List	Empty	Stores previous results

Implementation Snippet

```
var billInput by remember { mutableStateOf("") }
var numPeople by remember { mutableStateOf(1) }
var tipPercent by remember { mutableStateOf(15) }
var roundUp by remember { mutableStateOf(false) }

val result = calculateTip(billInput, tipPercent, numPeople, roundUp)
```

Test Cases

1. Valid input
   bill = 100, numPeople = 4, tip = 20% → correct split result.
2. Invalid input
   bill = -10 → error message displayed.
3. Edge case
   numPeople = 0 → show validation error.

Scaling Considerations

This feature can be extended by adding multiple currencies and persistent history using a local database or DataStore. Separating logic into functions and proper state management helps maintain scalability.

Reflection

The most complex part of this task was planning how different inputs affect the application state. Through this task, I gained a better understanding of state management and how Compose automatically updates UI based on state changes.